and precautions for using the CZ Series.
Before beginning operation, please read this manual carefully to get the most from your CZ Series.
Keep this manual handy for future reference.

RGB Digital Fiberoptic Sensors

CZ Series

Instruction Manual

- **The CZ Series is intended for the detection of target objects. Do not use the CZ Series in a safety circuit to protect the human body.**
- **The CZ Series does not have an explosion-proof structure. Do not use it in a location where any flammable gases, liquid or powder exist.**

ACCESSORIES

Instruction manual (This manual): 1

Mounting bracket: 1 Plastic screw driver: 1

SPECIFICATIONS

SAFETY PRECAUTIONS

This manual describes the instructions, operating procedures

Amplifier Fiber unit

Model
CZ-V1
Light source Response time
300 ∝s/1 ms (switch selectable)
Indicators
Output: Red LED, Calibration: Orange LED, External synchronization input: Green LED, Matching rate/received light intensity: LCD (Red/Green)
Error indication
Red LED, Green LED, Blue LED
Calibration method Tolerance value adjustment
selectable) Timer OFF/OFF-delay timer (40 ms) (switch selectable) Match output: Turns on when target color
Differentiation mode Timer function
Output mode selection
External synchro nization input
insufficient color difference
1-point/2-point calibration (switch selectable) Numerical value setting on digital display
C mode/C + I mode/I mode (switch
External calibration input
Registered color selection
matches registered color.
Mismatch output: Turns on when target color is different from registered color. (switch selectable)
Control output
Protection circuit
Input response time: 20 ms min.
8-bank selection (By external input or key operation), Input response time: 20 ms min.

Type
Reflective
Excess light intensity, insufficient light intensity,
Model
Detection range
Response speed: 500 ∝s max.
Minimum bend radius Enclosure rating
Ambient temperature Relative humidity
Fiber length
Housing material Weight
Detection method
R25 mm
IP40
–40 to +70°C (–40 to 158°F), No freezing
35 to 85%, No condensation
2 m (free-cut) 1 m (free-cut) Lens housing:
Reflective

Smallest spot diameter
Adjustable small beam spot
CZ-10
10 to 30 mm
0.9 to 3.5 mm dia.
Adjustable small beam spot, sideview
CZ-11
3 to 15 mm
0.9 to 1.5 mm dia.
Aluminum, Fiber case: Stainless steel
Approx. 5 g Approx. 13 g

NPN open-collector: 40 VDC max. (100 mA max.), Residual voltage: 1.0 V max.
Reverse-polarity protection (power supply), overcurrent protection (output), surge protection (output)

Type
Detection range
Smallest spot diameter
Minimum bend radius
Enclosure rating

mm
6 mm dia.
R25 mm

CZ-41
Small beam spot 16±4 mm
1 mm dia.
R15 mm

CZ-40
Long detecting distance 70±20

Model

IP67

Power supply Current consumption

Incandescent lamp: 5,000 lux max.,

10 to 55 Hz, 1.5 mm double amplitude in X, Y, and Z

35 to 85%, No condensation

Ambient light Ambient temperature [1] **Relative humidity**

12 to 24 VDC±10%, Ripple (P-P): 10% max. 75 mA max.

Sunlight: 10,000 lux max.

–10 to +55°C (14 to 131°F), No freezing
35 to 85%, No condensation

Ambient temperature Relative humidity Fiber length
Housing material Weight

2 m (free-cut)
Polyarylate
Approx. 27 g

–40 to +70°C (–40 to 158°F), No freezing

Vibration
Shock
Housing material Weight (including 2 m cable)

directions, 2 hours respectively
500 m/s² in X, Y, and Z directions, 3 times respectively

Polycarbonate

Approx. 115 g

1. When several units are connected, the acceptable ambient temperature varies depending on the conditions given below. To connect several units, be sure to mount them to a DIN rail (metallic plate). Ensure that the output current is 20 mA max.
• When 3 to 10 units are connected: -10 to +50°C (14 to 122°F) • When 11 to 16 units are connected: -10 to +45°C (14 to 113°F)

CONNECTING FIBER UNIT AND AMPLIFIER

1

1. Tilt the quick-release lever.

2. Push the single-core fiber to the transmitter side, and the multiple core fiber to the receiver side as far as they will go (Approx. 14 mm of the fiber will be inserted.).

* Inserting the fiber in the wrong side will decrease the original detection performance. Be sure to check the markings on the amplifier's lateral side before inserting the fiber.

3. Raise the quick-release lever.

1. Detach the end units.

2. Slide the expansion unit that is to be detached. Detach it individually from the DIN rail.

Note 1: When connecting several amplifiers, be sure to use a DIN rail and the end units.
Note 2: Be sure to turn the power off before connecting/disconnecting amplifiers.
Note 3: Do not remove the protective cover on the expansion connector from the outermost unit.
Note 4: Do not detach several units from the DIN rail while they are connected to each other.
Note 5: When several units are connected, confirm that the ambient temperature is appropriate. (See "Specifications" on page 1.)

1 3

■ DETACHING EXPANSION UNITS FROM DIN RAIL

Quick-release lever

Single-core fiber (with markings)

2

Transmitter side (with markings) Receiver side

MOUNTING FIBER UNIT

• Use the supplied special mounting bracket to mount the fiber unit in the desired position according to the location.

• Be sure to limit the tightening torque to 0.3 Nm or less.

Multiple-core fiber

Single-core
(Transmitter side)
Multiple-core (Receiver side)

MOUNTING AMPLIFIER

Mounting example 1 Mounting example 2

unhook the front claw by lifting the amplifier front side while pressing it

■ Mounting/detaching amplifier to/from DIN rail or mounting bracket

Hook the claw on the rear side of the amplifier onto the DIN rail or the mounting bracket, and then hook the front side claw to the rail or bracket while pressing the amplifier forward. To detach the amplifier, forward.

Mounting

Detaching

Reference: To cut the fiber to the desired length, use the special cutter included with the fiber unit.

CIRCUIT

INPUT/OUTPUT

Connections

Brown
100 mA max.

12 to 24 VDC

Yellow/purple

External bank selection 1

Green/purple

External bank selection 2

Blue

External bank selection 3 0 V

12 to 24 VDC

■ Side mounting

Using the side holes of the supplied mounting bracket, secure the amplifier with the screws.

1. Remove the protective cover on the side of the amplifier.

CONNECTING SEVERAL AMPLIFIERS

■ Mounting expansion units

Up to 16 expansion units (FS-T2, FS-M2, FS-V12, PS-T2) can be mounted to the side of the CZ-V1 amplifier.

Output circuit

Overcurrent protection

Black

Load Pink

Photoelectric sensor

Purple

Orange/purple

Brown

Black

5 to 40 VDC

External calibration External synchronization input

Load 5 to 40 VDC 100 mA max.

2. Mount expansion units to the DIN rail one at a time.

Remove the protective cover.

main circuit

3. Slide one expansion unit toward the main unit or another unit. Align the front claws of the units and push them together until you hear a click.

4. Secure the units together by pushing the end units (included in the expansion unit) from both sides.

Align the claw.
End unit (included with

Blue

External calibration input
0 V

Input circuit

Expansion units
the expansion unit)

External synchronization input External bank selection input 1 to 3

Main unit

included with the expansion unit. Attach the sticker

Photoelectric sensor

Orange/purple
Yellow/purple
Green/purple

main circuit
+5 V

PLC, etc.

Up to 16 expansion units can be connected.

+5 V Pink

0 V
Blue

Purple

The sticker on the right is

close to the sensor units.

Short-circuit current: Approx. 1 mA

2

PART NAMES

DIP switches for mode setting

HSPD FINE 2-P 1-P C+I C

I

High-speed/High-precision

2-point/1-point calibration

Differentiation mode selection

• Changes display.

UP/DOWN key
• Changes sensitivity setting value.
• Changes channels.

LCD display monitor

MODE selection button

Shows setting value.

SET button

4MODE 3

OUT TIM SET

CH

Output indicator

2

1

SET

• Shows matching rate. •

SETTING EACH

MODE

External synchronization input indicator

Timer selection switch
Switches between Timer OFF and 40 ms OFF-delay.

Output selection switch
Switches between N.O. and N.C.

Calibration indicator

DLY. OFF N.C N.O

Sets sensitivity.

4MODE

■ **FINE/HSPD selection (Using DIP switch 4)** Use HSPD when the detection requires a response speed less than 1

DIP switch

321

CH

ms.

SET

M h ode S n witc Descriptio

OUT TIM SET

DLY. OFF N.C N.O

✳

Timer selection switch Output selection switch

Factory setting

FINE
(Fain)
HSPD
(High-speed)

■ N.O./N.C. selection

Differentiates colors with high precision.
Differentiates colors with a high-speed response of 300 ∝s.

4
4

"✻" indicates the factory-set mode. Normally, you should use the CZ V1 with the setting indicated by "✻", and only change the setting if required.

Change the setting to invert the output mode. **M h ode S n witc Descriptio N.O.**

■ Differentiation mode setting (Using DIP switches 1 and 2)

Change the setting according to the detection conditions such as the target color or received light intensity.

✻ N.C. N.O.
(Color match output)
N.C.
Output is turned on when the target color matches the registered color.

Output is turned on when the target color does not match the registered color.

M h ode S n witc Descriptio

C (Color) Detects color using color components (R, G, and B) and received light intensity (received light quantity).
2

C + I
(Color and intensity)
1 2
Detects color using received light intensity (received light quantity).

✻
(Color mismatch output)
N.C. N.O.

I
(Intensity)
or
1 2

Detects color using color components (R, G, and B).

Sensitivity setting in C or C + I mode ➞ Go to page 4.
Sensitivity setting in I mode ➞ Go to page 5.

■ Timer OFF/40 ms OFF-delay selection

Change the setting to delay the output timing.

M h ode S n witc Descriptio

Timer OFF Output is turned on without any delay. ✻ **DLY. OFF**

40 ms OFF-delay Output is delayed for 40 ms. **DLY. OFF**

■ 1-point/2-point calibration selection[1] (Using DIP switch 3)

Change the calibration method.

M h ode S n witc Descriptio		
✻ **1-P (1-point calibration)**	Detects one specified color. (The sensitivity is automatically set to detect 3	Differentiates two specified colors. (The sensitivity is automatically set to 3 the optimal value to differentiate the two colors selected during calibration.)
2-P (2-point calibration)	only the one color selected during calibration.)	

1. The setting of DIP switch 3 is effective only in the C and C + I modes. The setting is unnecessary in I mode.

OPERATING PROCEDURE FOR USING C OR C + I MODE

Setting sensitivity

■ 1-point calibration (Detection of one specified color)

Place a target that is the reference color in the detection area of the fiber unit. Press the SET button and then release it.

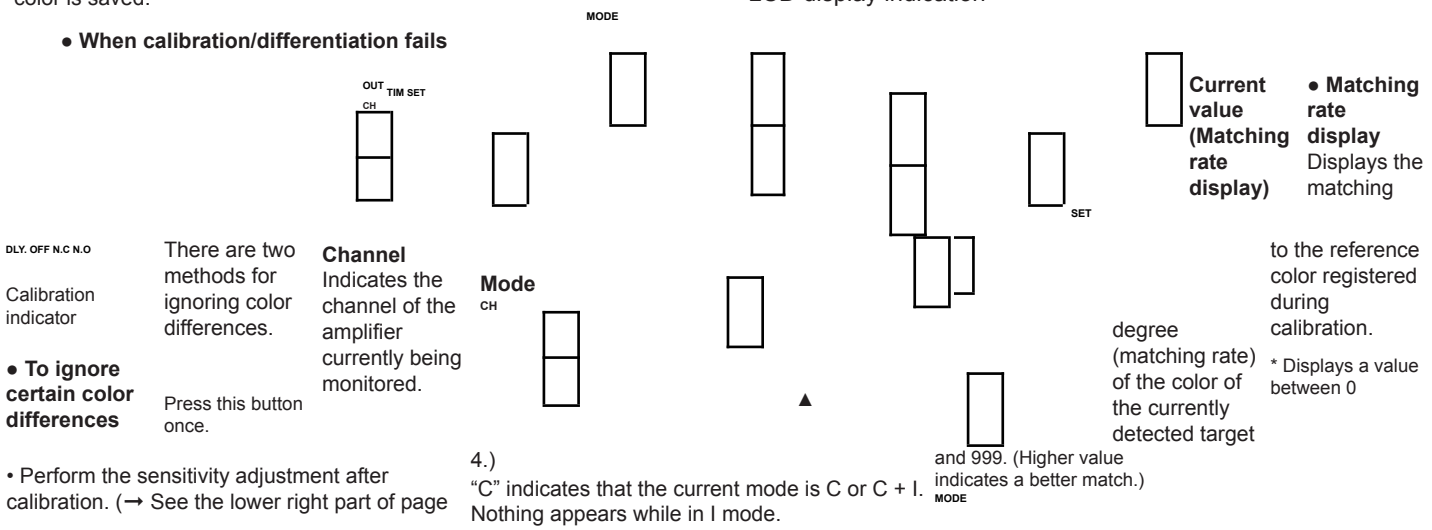
The calibration indicator illuminates momentarily and the reference color is saved.

Check the following points.

- Check whether the fiber unit is mounted properly (detection distance and detection angle). (See page 7.)
- Perform the sensitivity adjustment. (See below.)
- Select the other differentiation mode and then perform calibration again. (See page 3.)

LCD display indication

● When calibration/differentiation fails



● To ignore certain color differences

Press this button once.

- Perform the sensitivity adjustment after calibration. (➞ See the lower right part of page 4.)
- Use the quick color difference cancellation function. In 1-point calibration, hold down the SET button and continue the calibration with targets of different colors. The CZ Series ignores the color variation and saves them as the reference color.

"C" indicates that the current mode is C or C + I. Nothing appears while in I mode.

Press the button once.

▲

CH

OUT TIM SET

DLY. OFF N.C N.O

Calibration indicator
(Illuminates while the
SET button is held down.)

Setting value

MODE SET

CH

Hold down the SET button.
flashes on and off.

When "nnn" or "VVV" is
saved.

value of the
display
Displays
the
threshold
value
(criteria)

● **Setting**

The setting value
matching rate.

displayed
These displays indicate that the
received light is insufficient or
excessive. ➞ See page 7.

■ 2-point calibration (Differentiation of two colors)

1. Place a target that is the reference color in the detection area of the fiber unit. Press the SET button and then release it.

The calibration indicator illuminates and the reference color is

4 MODE
CH

3
2
1
OUT TIM SET

CH

DLY. OFF N.C N.O

Calibration indicator Press the
SET button
once.

2. Place a target that is the color to be differentiated in the detection area. Press the SET button and then release it.

Sensitivity adjustment

The tolerance of the detection can be adjusted.

1. Press the MODE button to show the setting value. MODE

SET

SET
M SET

Press
OFF N.C N.O

2. Press the UP/DOWN button to change the setting value.

Press

CH CH

The calibration indicator goes off. The CZ Series sets the optimal sensitivity to differentiate the two colors.

CH

OUT TIM SET
MODE SET

A higher value means precise detection. Press

A lower value means rough detection.

Calibration indicator Press the

Wide tolerance range

Narrow tolerance range

DLY. OFF N.C N.O

SET button once.

the LCD display monitor shows “- - -” for 3 seconds after calibration. In this case, see “When calibration/differentiation fails” on the upper right part of this page and try the calibration again.

* After the adjustment, press the MODE button to return to the matching rate display.

● **When “- - -” is displayed:**
When the sensitivity difference is insufficient for proper detection,

OPERATING PROCEDURE FOR USING I MODE

The display is factory-set to show the received light intensity. MODE

Setting sensitivity

Select the sensitivity setting procedure according to the target condi tions.

LCD display indication

■ To set sensitivity using a moving target (Fully-automatic calibration)

Excess gain (%) Hold this button down

4MODE
CH CH for 3 seconds or more.

Received light intensity (4 digits)

1. Pass a target through the optical axis while pressing the SET button (3 seconds or more).

3
2
1
OUT TIM SET

CH

SET

MODE MODE

2. Confirm that the calibration indicator (orange LED) flashes.

DLY. OFF N.C N.O

Press this button once.

Setting value (4 digits) CH

Press this button once.

3. Release the SET button. The calibration indicator (orange LED) goes off.

Calibration indicator

Press the button for 3 seconds or more.

Channel display: Indicates the channel of the amplifier currently being monitored.

■ Excess gain display

* The setting value flashes on and off.

CH

Received light intensity is converted by defining the setting value as 100P (%).

■ To detect a minute color difference (2-point calibration)

1. With a target in place, press the SET button and release it. The calibration indicator (orange LED) illuminates.

3
2
1
OUT TIM SET
CH

4MODE

• Stable LIGHT status is obtained with 110P (%) or more. • Stable DARK status is obtained with 90P (%) or less.

This display enables sensitivity adjustment using the UP/DOWN button.

SET

2. With the target removed, press the SET button and release it.

DLY. OFF N.C N.O

The calibration indicator (orange LED) goes off.

■ Received light intensity display

Received light intensity is displayed in 4 digit numbers by defining the maximum value as approximately 4000.

CH

depending on the fiber unit characteristics.

Calibration indicator

1. Press and release the button with a target in place.

2. Press and release the button without the target.

The maximum/minimum values vary

CH

■ Setting value display

(Positioning calibration)

The current setting value is displayed.

This display enables sensitivity adjustment using the UP/DOWN button.

■ For target positioning

4MODE

1. With no target, press the SET button and release it.

OUT TIM SET
CH

The calibration indicator (orange LED) illuminates.

3
2
1

setting value

SET

You can change the setting value in the "Setting value display" and "Excess gain display" modes.

Changing the

2. Place a target in the position where it is to be stopped.

DLY. OFF N.C N.O

Calibration indicator

Press the button and release it.

In setting value display mode

increases the sensitivity.

MODE

3. Press the SET button for 3 seconds or more and confirm that the calibration indicator (orange LED) flashes.

SET

4. Release the SET button.

MODE
SET

(The setting value is decreased.)

decreases the sensitivity.

■ For stable detection unaffected by dust or dirt (Maximum sensitivity setting)

(The setting value is increased.)

1. Under the conditions shown in the figure, press the SET button for 3 seconds or more until the

3
CH
2

4MODE

SET
1

In excess gain display mode

calibration indicator
(orange LED) flashes.

OUT TIM SET
DLY. OFF N.C N.O

MODE SET

increases the sensitivity.

Press the button

With no target

Calibration
indicator

more.

(The excess gain is

2. Release the SET button.
for 3 seconds or

MODE SET

increased.) decreases the sensitivity.

● **When the sensitivity difference is insufficient:**
"- - - - -" flashes for 3 seconds on the LCD display monitor after

(The excess gain is decreased.)

calibration.

UP/DOWN button
quickly changes
displayed values.

CH

5
Reference: Holding
down the

OTHER FUNCTIONS (COMMON TO C, C + I, AND I MODES)

Locking the sensitivity setting

Hold down the MODE button and press the UP button for 3 seconds.

monitor.

This locks the operation buttons.

colors from A through H.

• "Loc" flashes on and off on the LCD display
when in setting
locked status.

• You can still
change the display
mode and channel

4MODE

Cable color

Orange/
Yellow/

Green/
Blue

CH
Bank
purple
purple

purple
Green/purple

32

OUT TIM SET 2 1

1

SET

DRY. OFF N.C N.O

● **To cancel the locked status**
Repeat the procedure above while in
setting locked status. "UnL" flashes on
and

off on the LCD display monitor.

C X ● X **D** ● ● X **E** X X ● **F** ● X ● **G** X ● ● **H** ●●●
●: X:

● **Selecting registered colors (banks) using an external input**
The external bank selection cable allows the selection of 8
registered

+V
Brown

Orange/purple

Yellow/purple

0V

A XXX **B** ● X
X

Setting sensitivity using an external input (External calibration function)

1. Lock the sensitivity setting using the

Controlling expansion units (FS-T2, FS-V12, PS-T2) 1. When "CH"

is displayed, hold down the MODE button and momentar

procedure above. (If “UnL” is displayed,

repeat the same procedure.)

2. Connect the pink cable to a switch or a PLC.
The minimum input time is 20 ms.

Brown

ily press the UP/DOWN button. You can change the channel of the amplifier being controlled. Select the channel of the desired expan sion unit.

• The number of channels varies depending on the connected expansion units. Up to 16 expansion units can be connected.

NPN +V

• The channel number starts at 2, which is the number of the expan

Blue

sion unit next to the main unit, and it increases to 3 through to 17.

0 V

MODE

CH

+

Pink

3. The external calibration procedure is the same as using the SET button. (→ All the setting procedures on pages 4 and 5 are available

with the external input.)

Channel Main unit Expansion unit MODE

+

CH2 CH3 CH4 CH17

Selecting CH display

The CZ-V1 can internally store 8 colors (banks). The banks can be selected using the buttons or an external input.

● Selecting registered colors (banks) using the buttons

4MODE

1. Hold down the MODE button* and press the DOWN CH button for 3 seconds. “CH” appears and flashes on the LCD display.

OUT TIM SET

CH

3
2
1

indication

SET 1

Channel number

2. Hold down the MODE button* and momentarily press the UP/

DRY. OFF N.C N.O 2

4MODE

DOWN button. The banks are changed each time the UP/ DOWN button is pressed.

OUT TIM SET

CH

1-line expansion units

3

2

1

There are 8 banks from A

DRY. OFF N.C N.O

through H.

MODE +

condition of the selected expansion

unit.

2. You can use the CZ-V1 to

operate the selected expansion unit, for example to change the monitor display or sensitivity setting.

SET 1

* The red and green color of the LCD display represents the ON/OFF

2

Note 1: The FS-M2 cannot be controlled using

the CZ-V1.

Note 2: When the channel of the FS-M2 is selected, "- - - -" appears on

MODE +

External synchronization function

the LCD display. (The red or green color of the LCD display represents

the ON/OFF condition of the FS-M2.)

signal (connected to 0 V), the control output retains the condition at that exact moment.

3. To finish the selection, hold down the MODE button and press the DOWN button for 3 seconds.
"CH" stops flashing and remains illuminated.

Detection condition Match

4MODE

3

When the external synchronization input cable (purple) receives a

2

1

OUT TIM SET

CH

SET 1 CH

Control output

OFF ON

External Mismatch ON

DRY. OFF N.C N.O

2

→

synchronization input

OFF

*** Be sure to hold down the MODE button first when operating the MODE and UP/DOWN buttons.**

This output condition is retained.
This output condition is retained.

HINTS ON CORRECT USE

Mounting amplifier

- If the amplifier cable is placed together with power lines or high voltage lines in the same conduit, detection errors may occur due to noise interference. Isolate the amplifier cable from these lines.
- If there are several colors in a single beam spot, the CZ-V1 deter mines the color by averaging those colors. Therefore, it may produce an output even though the color is different from the registered color.
- To extend the cable length, use a cable with at least a 0.3 mm^2 nominal cross-section area. Limit the length of cable extension to 100 m or less. (To connect several units, contact Keyence for further information.)
- When using a commercially available switching regulator, ground the frame ground terminal and ground terminal.

- **Target movement and fiber unit orientation**
To stabilize the sensor output at a border, mount the fiber unit parallel to the border line as much as possible.

Incorrect Correct

- Do not use the CZ-V1 outdoors.
- Even when the same color is detected, the displayed value may vary depending on the characteristics of each amplifier, the length of the fiber cable and the location.

Fiber unit

Target movement direction

Target movement direction

- When any of the external inputs (calibration, synchronization or bank selection) are not used, cut the appropriate input cables (pink, purple, orange/purple, yellow/purple or green/purple) at the root or connect them to the "+" terminal of the power supply.

Mounting fiber unit

- A detection error may occur when the target is subjected to direct or reflected light from high-frequency lighting equipment such as an inverter fluorescent lamp. In such a case, apply a light shield plate or
- Prevent any objects from bumping the sensing surface.
- Do not pull the fiber cable of the CZ-40 and 41 using a strength of

greater than 20 N, 3 seconds.

ERROR INDICATIONS

The following indications on the LCD display show the error

change the location of the fiber unit.

Inverter fluorescent lamp, etc.

Light shield
plate

• **When detecting a metal surface or glossy target** When a target has a metal or glossy surface, the calibration/differentia tion may fail. To detect such a target, tilt the fiber unit by approximately 10 to 15 degrees.

10° to 15°

nnn

events. **Error**

indication C s ause Countermeasure

differentiation fails” on page 4 and retry the sensitivity adjustment.

vvv

- - - -
Received light intensity is insufficient.

Received light intensity is excessive.

Color difference (in I mode, “difference in light intensity”) is too small to differentiate colors during 2-point calibration.

Check whether the fiber unit is installed at the specified detecting distance. Insert the fiber cable into the fiber unit as far as it will go (approx. 14 mm).

Tilt the fiber unit by approx. 10 to 15 degrees. (→ See “Mounting fiber unit” on the left part of this page.)

→ See “When calibration/

Correct the problem using the following

countermeasures. **DIMENSIONS**

CZ-V1

When mounted to DIN rail

ø5.2 x
Brown, blue: 0.45 mm²
Black, Pink, Purple, Orange/purple,

green/purple, yellow/purple: 0.25 mm²

When mounting bracket is

attached 1.1

2-ø3.2

When the cover is opened: 78max

5.8

Cable length: 2 m

36.5

5

11.6

18.5 15

8

24 min.

16

24

6.7

8.4

6.5 35.4

Amplifier mounting bracket (accessory) 2 - (4.4 x 3.4)

10.7

24 min.

(12.7)

22

CZ-11 2-ø3.2

ø4.5 stainless steel

ø3.5 stainless steel

15

28.5

SPCC

CZ-10

53.5

ø6 A M6 P=0.75 mounting part

2-ø1.3 transmitter

5.2

A-A

cross

ø5.5 lens 2.6

3 A

10 9.6

L 2000 *

45.0 10.4 L 5.0 8.7

2.1

ø4.5 stainless steel 5.0

section view

L: lens unit length *adjustable part

4.2 to 9.3

5.5 15.0 12.0

2.4 5.4 5.6

300

1000

M3, P=0.5

lens ø3.7

13.2

25.5

13.2

27

CZ-40

5.8

6.3

13.2

Slot (Note)

CZ-41

5.8

6.3

Lens housing: Black anodized aluminum

L: 2.6 to 5.0

13.2

14 11.5

28

33 2000

17.5 11.5

32

37 2000

0.5 11.5

(Note) Detail of slot

ø3.3 ø2.2 x 2 (Mounting hole)

2 - ø3.3 ø2.2 x 2

(Mounting hole)

28.4

(accessory)

3.6 9.5 **Detail of slot** 2 x M3 screw

When mounting bracket is attached to CZ-40

Fiber unit mounting bracket

5

3.5 4

3.5

6

17.5

8

Stainless steel

11.5

3.6

11.5

14.5

45

33

11.5

18

t = 1.2

t = 1.5

(32)

Stainless steel

The fiber unit angle can be changed by a maximum of 45 degrees upward and downward.

18.7

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